Necessity, Reasoning, and Position: Complementarity-Aware Evidence Fusion for Multimodal Emotion-Cause Pair Extraction in Conversations

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***Abstract*—Multimodal emotion-cause pair extraction (MECPE) aims to identify, for every utterance that expresses an emotion in a conversation, the utterances that cause that emotion, using textual, acoustic, and visual signals together with speaker identity. Despite rapid progress, existing models share two weaknesses. First, their pair decisions are vulnerable to *spurious correlation*: a candidate is predicted as a cause without ever testing whether removing it would actually change the prediction, so the model may latch onto a dominant modality or a co-occurring topic. Second, causes exhibit strong *positional regularity*, which is either ignored or allowed to override semantics, producing position-biased predictions. We propose a model that augments a relation-aware multimodal encoder with three complementary sources of decision evidence: (i) perturbation-based *necessity evidence* that probes how much each candidate cause and each non-verbal modality contributes to a pair prediction; (ii) a *bounded positional prior* that exploits the locality of causes without letting position dominate; and (iii) *reasoning evidence* distilled offline from a large language model (LLM) as text-grounded counterfactual judgements. A complementarity-aware gating network fuses these signals conditioned on model confidence, teacher reliability, and their agreement, so each source compensates the others' weaknesses; the LLM is queried only offline and never at inference. On the benchmark Emotion-Cause-in-Friends (ECF) corpus the method improves the emotion-cause pair F_1 over strong baselines, and extensive ablations confirm that every component contributes.**

***Index Terms*—Affective computing, emotion-cause pair extraction, multimodal conversation understanding, graph neural networks, counterfactual reasoning, knowledge distillation.**

I. Introduction

UNDERSTANDING *why* a speaker feels a certain way is central to affective computing. Beyond recognizing that an utterance conveys an emotion, many applications—empathetic dialogue systems, opinion mining, and mental-health support—require identifying the events or utterances that *trigger* that emotion. Emotion-cause extraction (ECE) was first framed as detecting the clauses that cause a given emotion [1]; emotion-cause pair extraction (ECPE) then removed the need for emotion annotations at test time by jointly extracting emotion and cause clauses as pairs [2]. In conversations, emotions and their causes are conveyed not only through text but also through tone of voice and facial expression, which motivates the multimodal emotion-cause pair extraction (MECPE) task [3]: for each emotion-bearing utterance in a multi-party conversation, find the utterances that cause it, using text, audio, and video.

The conversational and multimodal setting is precisely what makes the task valuable in practice. In an empathetic dialogue agent, knowing that a user's frustration was caused by an earlier system response, rather than by the current utterance, changes how the agent should repair the interaction. In opinion mining over multimedia content, attributing a sentiment to the event that triggered it—spoken with a particular tone or accompanied by a particular facial expression—yields far more actionable insight than a sentiment label alone. These applications require not only *accurate* pair predictions but also predictions that are *robust* to superficial cues and, ideally, *explainable*, because an attribution that cannot be justified is difficult to act upon.

MECPE is challenging for several intertwined reasons. The label space is quadratic in the number of utterances and highly imbalanced, since only a small fraction of utterance pairs are genuine emotion-cause pairs. A large share of causes are *self-contained*—the emotion is triggered by the event described *within* the emotion utterance itself, so the cause coincides with the emotion utterance. And causes exhibit strong but not absolute positional regularity: they tend to lie close to the emotion utterance, which is informative yet dangerous to over-trust.

Recent models substantially improve MECPE by modeling rich inter-utterance interaction with graph or transformer architectures and by injecting label constraints [4], and the latest work explores multi-LLM debate to reconcile divergent predictions [5]. However, two limitations persist and motivate this work (Fig. 1).

(1) Spurious correlation. Pair decisions are not grounded in any explicit notion of *necessity*. The model predicts a pair without ever asking whether removing the candidate cause—or a particular modality of it—would actually change the prediction. As a result it can rely on a co-occurring topic or on a single dominant modality, which inflates false positives and makes decisions hard to calibrate and to explain.

(2) Positional bias. Because causes cluster near the emotion utterance, a model that does not separate genuine semantic evidence from positional regularity tends to mechanically pair adjacent utterances, degrading precision when the true cause

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Fig. 1. Two failure modes that motivate this work. *Top:* spurious correlation—the model predicts a pair from a co-occurring topic or a dominant modality without testing whether the candidate cause is *necessary*. *Bottom:* positional bias—a nearby utterance is mechanically taken as the cause. Our model counters both with explicit necessity evidence and a bounded positional prior. [TODO: draw the real figure]

is distant or self-contained.

Both failure modes are instances of *shortcut learning*, in which a model exploits statistically predictive but non-causal features that fail to transfer to harder cases [6]. Our central design choice is therefore to make the model's pair decisions answerable to an explicit notion of necessity, rather than to whatever correlations happen to be predictive on the training distribution.

A tempting remedy is to query an LLM, which exhibits strong commonsense causal reasoning, for every candidate pair. This is impractical at the quadratic scale of MECPE, and an LLM cannot perceive the non-verbal modalities. We therefore restrict the LLM to *text-grounded* reasoning, query it *only offline*, and *distill* its judgements into a lightweight student that runs without any LLM call at inference.

Concretely, we augment a relation-aware multimodal encoder with three complementary sources of decision evidence. **(i) Necessity evidence** (Section III-D) is a model-internal, perturbation-based signal: we replace the candidate cause—and, separately, each non-verbal modality of the cause utterance—with a learned, data-anchored neutral baseline, re-score the pair, and measure the resulting drop, with a direct calibration loss anchoring the signal to the pair labels. **(ii) A bounded positional prior** (Section III-E) encodes relative position as a capped additive term on the pair score, regularized so position can refine but never override the semantic evidence. **(iii) Reasoning evidence** (Section III-F) is distilled offline from an LLM: for hard pairs, a structured prompt elicits text-grounded *counterfactual* judgements (necessity, sufficiency, direction, spuriousness) plus a self-agreement reliability estimate; a student then predicts these scalars from the pair representation alone. The three signals are reconciled by a **complementarity-aware gating network** (Section III-G) that conditions on the model's confidence, the teacher's reliability, and the agreement between the model's own necessity signal and the LLM's necessity judgement.

Our contributions are summarized as follows.

- We introduce a unified MECPE model that grounds pair decisions in explicit *necessity evidence*, obtained by faithful, pair-local perturbations of the cause utterance and of each non-verbal modality, and calibrated directly against the pair labels, directly attacking spurious correlation.
- We distill commonsense *counterfactual reasoning* from an LLM into a lightweight student, with a dedicated treatment of self-contained causes, so that semantically informed external knowledge is available at inference *without* any LLM call.
- We propose a *complementarity-aware fusion* mechanism, together with a bounded positional prior and a necessity-modulated agreement term, that lets each evidence source compensate the others' weaknesses under explicit reliability and agreement conditioning.
- On the benchmark ECF corpus the method improves emotion-cause pair F_1 over strong baselines, and extensive ablations isolate the contribution of each component.

The remainder of this paper is organized as follows. Section II reviews emotion-cause analysis, multimodal conversational modeling, interpretability and counterfactual reasoning, and LLM distillation. Section III formalizes the task and details the encoder, the three evidence sources, the fusion module, and the two-phase training procedure. Section IV presents the experimental setup and results, including the main comparison, ablations, modality and sensitivity analyses, and qualitative studies. Section V discusses why the evidence sources are complementary and states the limitations, and Section VI concludes.

II. Related Work

A. Emotion Cause Analysis

ECE was first framed as detecting the clauses that cause a given emotion [1], and multi-task formulations later coupled it with related emotion-analysis objectives [7]. Xia and Ding [2] introduced ECPE, which jointly extracts emotion and cause clauses as pairs without requiring emotion annotations at test time. A rich line of discriminative methods followed: two-dimensional representations [8], sliding-window multi-label learning [9], transition-based parsing [10], and effective inter-clause ranking [11]. A complementary line of generative methods reframes pair extraction as sequence generation or masked prediction, directly emitting descriptions of the emotion-cause relationship rather than classifying candidate clauses. Both families operate on text alone and on monologue-style documents, and the discriminative family in particular identifies causes by modeling characteristic correlations between clauses—precisely the setting in which spurious correlation can arise, since a clause that merely co-occurs with the emotion is easily mistaken for its cause. Relative position has long been recognized as a strong but double-edged cue for cause identification [12], which directly motivates our *bounded* treatment of positional information. Our perturbation-based necessity evidence is motivated by exactly this failure mode: rather than trusting a learned correlation, we test whether removing a candidate actually changes the prediction.

B. Multimodal Emotion-Cause Pair Extraction

Wang et al. [3] extended the task to conversations and to multiple modalities, releasing the Emotion-Cause-in-Friends

(ECF) corpus and establishing the MECPE task. Subsequent work advanced the state of the art with holistic interaction and label constraints (HiLo) [4], unified emotion recognition and cause modeling (UniMEEC) [13], multimodal heterogeneous graphs (M3HG) [14], and adaptive hierarchical reasoning (AHMRC) [15]. Most recently, DEC-Debate [5] exploits discrepancies between discriminative and generative predictions through a multi-LLM debate. Our work builds on the relation-aware encoder common to this line but departs from it by grounding pair decisions in explicit necessity evidence and by distilling external counterfactual reasoning offline, rather than relying on interaction modeling or on online LLM queries. Two characteristics of conversational MECPE shape our design and recur across this literature: a large fraction of causes are *self-contained* (the trigger lies within the emotion utterance), and genuine causes cluster near the emotion utterance. Prior models either treat the self-cause case implicitly or let positional cues enter the representation unconstrained; we instead give self-contained causes an explicit marker and bound the influence of position, as detailed in Section III.

C. Graph and Multimodal Modeling of Conversations

Conversational affect modeling has a substantial history, from recurrent context modeling [16] and contextual reasoning networks [17] to the broader study of emotion recognition in conversation [18], [19]. Graph neural networks are a natural fit for conversations, where utterances relate through speaker identity and reply structure [20]. We use relation-specific attention in the spirit of relational GCNs [21] and graph attention networks [22], and we incorporate commonsense context in the spirit of [23]. For multimodal fusion, prior work disentangles modality-invariant and modality-specific factors [24], fuses modalities with tensor products [25], and aligns modalities with cross-modal transformers [26] built on the attention mechanism [27]; we keep the fusion light and instead focus the modeling effort on causal-style evidence. A recurring concern in conversational modeling is the gap between training, where gold emotion labels can shape the graph, and inference, where those labels are predicted; we address this directly with scheduled sampling on the emotion-transition relation (Section III-B), so the training-time graph converges to the inference-time graph rather than relying on an idealized structure that is unavailable at test time.

D. Interpretability and Counterfactual Reasoning

Perturbation-based attribution explains predictions by measuring sensitivity to input changes [28], [29]. We repurpose this idea *inside* training: necessity evidence is a model-level signal that both regularizes and explains pair decisions, distinct from asserting causal relations in the world. Counterfactual reasoning has deep roots in causal inference [30] and has been used to debias vision-language [31] and relation-extraction [32] models; mitigating spurious correlation is a recognized challenge in machine learning [33]. The closely related task of jointly extracting multimodal emotion-cause pairs has also been studied in earlier multimedia work [34]. We obtain weak, text-grounded counterfactual judgements by prompting an LLM, which is complementary to—not a substitute for—the model-internal perturbation signal. The two notions of "necessity" we use are deliberately different in kind: the perturbation signal is faithful to the model but blind to world knowledge, whereas the LLM judgement carries world knowledge but is only loosely tied to the model; their agreement is therefore informative, and we exploit it explicitly in the fusion gate (Section III-G). Mitigating positional shortcuts also connects our work to the broader effort to suppress dataset-specific shortcuts in structured prediction [33].

E. LLMs and Knowledge Distillation

LLMs exhibit strong commonsense reasoning when prompted to reason step by step [35], and recent open models [36]–[38] make such reasoning broadly accessible, often adapted efficiently with low-rank adaptation [39]. Querying them at the quadratic scale of MECPE, and for non-verbal signals, is nonetheless impractical. We therefore distill their reasoning offline into a small student [40], in the spirit of symbolic knowledge distillation that compiles LLM commonsense into compact models [41], restricting the LLM to text-grounded judgements and obtaining modality evidence from perturbations instead. Compared with online debate-style use of multiple LLMs [5], offline distillation incurs a one-time annotation cost and adds *zero* LLM calls at inference, which is essential at the quadratic candidate scale of MECPE; the price is that the student can only approximate the teacher, a trade-off we examine empirically in Section IV. To control annotation cost without biasing the student toward easy pairs, we annotate a stratified mixture of hard and sampled easy pairs drawn from the same population the model sees at inference.

III. Methodology

We address MECPE in conversations, where the goal is to identify, for each utterance that expresses emotion, the utterances that cause it. Our model couples a relation-aware multimodal encoder (Section III-B) with three complementary sources of decision evidence: necessity evidence (Section III-D), a bounded positional prior (Section III-E), and reasoning evidence distilled offline from an LLM (Section III-F). The signals are combined by a gated fusion module (Section III-G) and trained jointly (Section III-H). Fig. 2 gives an overview.

A. Task Formulation

A conversation is a sequence of N utterances $\mathcal{D} = \{u_1, \dots, u_N\}$. Each utterance carries textual, acoustic, and visual features together with a speaker identity, $u_i = \{x_i^t, x_i^a, x_i^v, s_i\}$. For an ordered pair (u_i, u_j) in which u_i is the candidate *emotion* utterance and u_j the candidate *cause* utterance, the model predicts

$$p_{ij} = P(y_{ij} = 1 \mid u_i, u_j, \mathcal{D}) \,, \tag{1}$$

where $y_{ij} = 1$ iff u_j is a cause of the emotion expressed in u_i. Two auxiliary utterance-level heads predict an emotion probability $\hat{p}_i^e$ and a cause probability $\hat{p}_j^c$; these supply supervision and drive the candidate gating of Section III-C. We

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Fig. 2. Overview of the proposed model. A relation-aware multimodal encoder produces contextualized utterance and pair representations; a soft candidate gate selects the pairs that receive evidence; perturbation-based necessity evidence, a bounded positional prior, and distilled counterfactual reasoning are fused by a complementarity-aware gate to predict the pair probability. The LLM is used *only* offline to produce distillation targets. [TODO: draw the real architecture figure (adapt the diagram in the `Graph/` folder)]

adopt the standard convention that a cause utterance does not follow its emotion utterance, restricting candidates to $j \le i$, which yields the upper-triangular region used for negative sampling in Section III-H. The self-pair (i, i) is retained as a first-class candidate because, as noted above, a large fraction of emotions are triggered within the emotion utterance itself; treating such pairs as a special case rather than discarding them is a recurring theme in our design.

B. Relation-Aware Multimodal Encoder

The encoder produces context-aware utterance representations whose interactions reflect both the conversational structure (who speaks to whom, and in what order) and the affective dynamics (where emotions shift). We deliberately keep the multimodal fusion simple and place the modeling emphasis on the relation-aware propagation and on the downstream evidence, so that the gains we report can be attributed to the evidence pathway rather than to an elaborate fusion module.

Multimodal fusion. Each utterance is projected into a shared H-dimensional space by fusing its modalities and speaker identity, $h_i^0 = \mathrm{Fuse}(x_i^t, x_i^a, x_i^v, s_i) \in \mathbb{R}^H$.

Conversation graph. We build a heterogeneous graph over utterances with four relation types $\mathcal{R} = \{$*self-loop*, *same-speaker*, *reply/adjacent*, *emotion-transition*$\}$: *self-loop* ($i = j$); *same-speaker* ($s_i = s_j, i \neq j$); *reply/adjacent* ($|i - j| = 1$ or u_j is the reply target of u_i, with a fallback to the nearest preceding cross-speaker utterance); and *emotion-transition*, where structurally connected turns whose emotion labels differ are re-typed, taking priority over the structural off-diagonal types. The emotion-transition relation depends on emotion labels, which are gold during training and predicted at inference. To avoid a train–inference mismatch, the source label of each such edge is drawn by scheduled sampling [42],

$$\ell_i = \begin{cases} \ell_i^{\text{gold}}, & \text{with probability } \varepsilon_t, \\ \ell_i^{\text{pred}}, & \text{with probability } 1 - \varepsilon_t, \end{cases} \qquad \varepsilon_t : 1 \to 0, \tag{2}$$

where ℓ_i^{pred} is the detached arg-max of a preliminary emotion head. As ε_t anneals to 0, the training-time graph converges to the inference-time graph. This relation and the evidence modules of Sections III-D and III-F are activated only after a warmup of K epochs.

Relation-aware propagation. Messages are aggregated with relation-specific attention,

$$\alpha_{ij}^r = \operatorname*{softmax}_j \Big(\mathrm{LeakyReLU}\big(a_r^\top [\, W_r h_i^{(l)};\, W_r h_j^{(l)} \,] \big) \Big), \tag{3}$$

$$h_i^{(l+1)} = \sigma\Big(\textstyle\sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_i^r} \alpha_{ij}^r \, W_r h_j^{(l)} \Big). \tag{4}$$

After L layers we obtain contextualized representations $\tilde{h}_i \in \mathbb{R}^H$, written compactly as $\tilde{H} = \mathrm{Enc}(\{h_i^0\})$. A candidate-pair representation concatenates the two utterance vectors with their element-wise product and absolute difference,

$$h_{ij}^{\text{pair}} = \big[\tilde{h}_i;\, \tilde{h}_j;\, \tilde{h}_i \odot \tilde{h}_j;\, |\tilde{h}_i - \tilde{h}_j| \big] \in \mathbb{R}^{4H}. \tag{5}$$

Self-contained pairs. For a self-cause pair ($j = i$) this representation degenerates: the difference block $|\tilde{h}_i - \tilde{h}_i|$ is identically zero and the product collapses to $\tilde{h}_i \odot \tilde{h}_i$, so a quarter of h_{ii}^{pair} carries no signal—an avoidable handicap given that such pairs are about half of the labels. We inject a learnable self-loop signature $e^{\text{self}} \in \mathbb{R}^H$ into the otherwise-dead difference block on the diagonal, $h_{ii}^{\text{pair}} = [\tilde{h}_i;\, \tilde{h}_i;\, \tilde{h}_i \odot \tilde{h}_i;\, e^{\text{self}}]$, which gives self-cause pairs a distinct marker while leaving the $4H$ dimension and every cross-utterance pair unchanged. Initialized at $e^{\text{self}} = 0$, it recovers the plain representation at the start of training, and is applied identically in the factual and counterfactual scorings of Section III-D, so the necessity difference is uncontaminated by the marker.

C. Soft Candidate Gating

Enumerating all pairs is quadratic and computing evidence for every pair is costly. We define a soft relevance gate from the auxiliary heads, $\pi_{ij} = \hat{p}_i^e \cdot \hat{p}_j^c \in [0, 1]$, and allocate the evidence computations to the top-M pairs by π_{ij}, denoted

$\mathcal{T}$. Gating governs *where computation is spent*, not which pairs may be positive: every valid pair still contributes to the pair loss, so no candidate is permanently discarded. This distinction is important for two reasons. It keeps the expensive evidence computation on the small set of plausible candidates, which is what makes the per-pair cost a constant $O(M)$ rather than $O(N^2)$ (Section III-I), and it ensures that the gate cannot become a hard bottleneck that silently caps recall: a pair that the auxiliary heads underrate still receives the full pair loss and can be predicted positive, it merely does not receive the auxiliary evidence. The presence weight introduced in Section III-G further smooths the boundary of $\mathcal{T}$, so the transition between "inside" and "outside" the budget is continuous rather than a hard cutoff that the optimizer could exploit.

D. Perturbation-Based Necessity Evidence

A candidate cause should matter only if removing it changes the prediction—the direct antidote to spurious correlation. The intuition is the counterfactual one familiar from attribution methods: if the model's score is unchanged after we erase the candidate cause, then the model was not really using it, and a high probability assigned to such a pair is suspect. We turn this intuition into a trainable, model-internal signal by performing *faithful* perturbations—ones that actually propagate through the encoder—and by calibrating the resulting score drops against the pair labels, so that the magnitude of the drop becomes a meaningful feature for the classifier rather than a post-hoc explanation only. A scalar pair-scoring head $f : \mathbb{R}^{4H} \to \mathbb{R}$ produces the factual score $s_{ij} = f(h_{ij}^{\mathrm{pair}})$, supervised by the pair labels, $\mathcal{L}_{\mathrm{score}} = \mathrm{BCE}(\sigma(s_{ij}), y_{ij})$.

Anchored baselines. Perturbations replace an element with a neutral baseline rather than zero. With detached exponential-moving-average means $\bar{h}$ (utterance space) and $\bar{x}^m$ (modality-m input space),

$$b^{\mathrm{utt}} = \bar{h} + \delta^{\mathrm{utt}},\ b^m = \bar{x}^m + \delta^m,\ \mathcal{L}_{\mathrm{anc}} = \|\delta^{\mathrm{utt}}\|_2^2 + \textstyle\sum_m \|\delta^m\|_2^2, \tag{6}$$

where the learnable residuals δ are penalized so the baselines remain near genuine "no-information" means.

Cause necessity. Replacing the contextualized cause representation $\tilde{h}_j$ by b^{utt} and re-scoring yields

$$s_{ij}^{-u} = f\Big(\big[\tilde{h}_i;\, \tilde{h}_j^{-u};\, \tilde{h}_i \odot \tilde{h}_j^{-u};\, |\tilde{h}_i - \tilde{h}_j^{-u}|\big]\Big),\ \Delta_{ij}^{u} = s_{ij} - s_{ij}^{-u}. \tag{7}$$

Because the textual stream is the backbone of the encoder, this representation-level term also captures the semantic necessity of the cause; we reserve modality perturbation for the non-verbal streams.

Pair-local modality contribution. For $m \in \{a, v\}$ a faithful perturbation must propagate through the non-linear encoder. We re-encode the whole conversation once per modality with that modality replaced by its baseline at the input, $\tilde{H}^{-m} = \mathrm{Enc}(\{h_i^0\} \mid x^m \leftarrow b^m)$, costing two extra encodings per conversation shared by all of its pairs. Keeping the original emotion representation and using the perturbed cause representation $\tilde{h}_j^{-m}$,

$$\Delta_{ij}^{m} = s_{ij} - f\Big(\big[\tilde{h}_i;\, \tilde{h}_j^{-m};\, \tilde{h}_i \odot \tilde{h}_j^{-m};\, |\tilde{h}_i - \tilde{h}_j^{-m}|\big]\Big), \tag{8}$$

which makes the modality term pair-local and directly comparable to the cause term. The normalized terms and modality weights are $\tilde{\Delta}_{ij}^{m} = (\Delta_{ij}^{m} - \mu_m)/(\sigma_m + \epsilon)$ and $w_{ij}^{m} = \mathrm{softmax}_{m \in \{a,v\}}(\tilde{\Delta}_{ij}^{m})$, with running statistics (μ_m, σ_m); the cause term is standardized within its own group as $\tilde{\Delta}_{ij}^{u} = (\Delta_{ij}^{u} - \mu_u)/(\sigma_u + \epsilon)$.

Necessity-evidence vector. Collecting the terms gives

$$z_{ij}^{\mathrm{nec}} = \big[\, \tilde{\Delta}_{ij}^{u};\ \tilde{\Delta}_{ij}^{a};\ \tilde{\Delta}_{ij}^{v};\ w_{ij}^{a};\ w_{ij}^{v} \,\big] \in \mathbb{R}^5, \tag{9}$$

consumed by the fusion module with a stop-gradient so noisy early evidence informs the classifier without destabilizing the backbone.

Necessity calibration. To ensure the necessity evidence is semantically meaningful—removing a true cause decreases the score, removing a non-cause does not—we add a hinge-style loss,

$$\begin{aligned}\mathcal{L}_{\mathrm{cal}} = \frac{1}{|\mathcal{B}|} \sum_{(i,j)\in\mathcal{B}} \Big[&\mathbb{1}[y_{ij} = 1]\,\mathrm{ReLU}(-\Delta_{ij}^{u}) \\ &+ \mathbb{1}[y_{ij} = 0]\,\mathrm{ReLU}(\Delta_{ij}^{u} - \kappa_{\mathrm{cal}})\Big], \end{aligned} \tag{10}$$

with a small margin $\kappa_{\mathrm{cal}} \geq 0$. The two terms encode the desired behavior directly: for a true pair the cause-removal drop Δ_{ij}^{u} should be positive (removing the cause hurts the score), and for a negative pair the drop should stay below the margin κ_{cal} (removing a non-cause should not look like removing a cause). Calibrating internal signals to be meaningful is a recurring theme in reliable prediction [43]. Without this calibration the perturbation magnitudes would be arbitrary and the classifier could not read them as evidence; with it, the magnitude of the drop acquires a consistent meaning across pairs, which is what allows the fusion module to use z_{ij}^{nec} as a feature rather than a mere diagnostic. We stress that this necessity evidence is a *model-level* quantity: it measures the sensitivity of the model's own prediction to controlled input perturbations, used to regularize and explain predictions rather than to assert causal relations in the world. The stop-gradient on z_{ij}^{nec} is deliberate: the evidence is allowed to inform the classifier, but gradients from the classifier are not allowed to reshape the encoder *through* the evidence, which would let the model game the perturbation by learning representations that produce convenient drops rather than faithful ones.

E. Bounded Positional Prior

Causes tend to lie near the emotion utterance, so relative position is informative; the risk is over-reliance. With normalized signed distance $d_{ij} = (i - j)/N \in [-1, 1]$ and a small network ψ,

$$b_{ij}^{\mathrm{pos}} = \eta \cdot \tanh(\psi(d_{ij})) \in (-\eta, \eta),\ \mathcal{L}_{\mathrm{pos}} = \frac{1}{|\mathcal{B}|} \sum_{(i,j)\in\mathcal{B}} \big(b_{ij}^{\mathrm{pos}}\big)^2. \tag{11}$$

The amplitude η caps the prior and $\mathcal{L}_{\mathrm{pos}}$ discourages it from dominating, so position can refine but not override the semantic evidence, mitigating positional bias. The design encodes a specific belief: position is a useful tie-breaker among semantically plausible candidates, but it must never be the sole reason a pair is predicted. The additive, bounded form makes this auditable—the maximum contribution of position to any pair score is η, which is small relative to the dynamic range of the semantic score—and the quadratic penalty pulls the prior toward zero unless the data provide consistent evidence that position helps. We verify in Section IV that increasing η beyond a small value re-introduces the positional shortcut and lowers precision, confirming that the bound, rather than the mere presence of a positional feature, is what matters.

F. LLM-Guided Counterfactual Evidence Distillation

LLMs reason effectively over text but cannot perceive raw acoustic or visual signals; we restrict their role to text-grounded reasoning and obtain modality evidence solely from Section III-D. Because querying an LLM at inference is impractical, its reasoning is distilled offline into a lightweight student. The motivation for adding an external teacher at all is that the model-internal necessity signal, while faithful, is blind to world knowledge: deciding whether an insult plausibly causes sadness, or whether two utterances merely share a topic, often requires commonsense the encoder has not learned from the limited training data. An LLM supplies exactly this commonsense, but only over text and only unreliably, which is why we (i) confine it to a small set of calibrated counterfactual scalars, (ii) attach a reliability estimate to each annotation, and (iii) let the fusion gate, not the LLM, decide how much to trust the resulting evidence for each pair.

Counterfactual reasoning evidence. For a candidate pair, a fixed structured prompt asks the LLM to reason counterfactually about an intervention on the candidate cause and to return four calibrated scalars in $[0,1]$: necessity s_{ij}^{nec} (how likely the emotion in u_i would disappear or weaken had u_j not occurred), sufficiency s_{ij}^{suf} (how likely u_j alone elicits the emotion), direction s_{ij}^{dir} (whether u_j causes u_i rather than the reverse), and spuriousness s_{ij}^{spur} (probability that u_i, u_j merely co-occur). Reliability is the inter-sample agreement $\rho_{ij} \in [0,1]$ across k stochastic samples. The reasoning-evidence vector is

$$z_{ij}^{\mathrm{rea}} = \left[\, s_{ij}^{\mathrm{nec}};\ s_{ij}^{\mathrm{suf}};\ s_{ij}^{\mathrm{dir}};\ s_{ij}^{\mathrm{spur}};\ \rho_{ij} \,\right] \in [0,1]^5. \tag{12}$$

These are *textual* counterfactual judgements obtained by prompting, not a formal structural-causal-model identification.

Self-contained causes. For self-cause pairs ($j = i$) the "remove u_j" framing is ill-posed, since deleting the utterance also deletes the emotion it carries. We branch the prompt on whether $j = i$: for $j \neq i$ the four scalars retain the cross-utterance reading; for $j = i$ the counterfactual is taken over the *event described* in u_i rather than the sentence. Both branches share the same four-scalar schema, so all downstream consumers are unchanged, and the self-pair (i,i) is always included among the annotation candidates for every non-neutral emotion utterance. This event-level reframing matters because self-contained causes are common in conversation—a speaker narrates an event and reacts to it in the same turn—and naively asking "would the emotion change if this utterance were removed" is degenerate, since the utterance *is* the emotion. By intervening on the described event instead of the surface sentence, the teacher can still produce a meaningful

System. You are an expert in conversational affect. Reason counterfactually and answer *only* with the requested JSON.
Context. Speakers and utterances $u_{i-w}, \ldots, u_i$ (with the emotion of u_i), the candidate cause u_j, and whether $j = i$ (self-contained).
Question. Consider intervening on u_j. If $j \neq i$, imagine u_j had not occurred; if $j = i$, imagine the event *described* in u_i had not occurred. Estimate, each in $[0,1]$: `necessity`, `sufficiency`, `direction` ($u_j \to u_i$ vs. reverse), `spuriousness` (mere co-occurrence).
Output. `{"necessity":_, "sufficiency":_, "direction":_, "spuriousness":_}`

Fig. 3. Structured prompt schema used to elicit text-grounded counterfactual judgements offline. Reliability ρ_{ij} is the agreement across k stochastic samples and is not produced by a single call. [TODO: paste the final prompt verbatim]

necessity judgement (would the speaker still feel this way had the event not happened), and the student inherits a coherent target for the self-pair. We found this distinction important in preliminary inspection: without it, the teacher's annotations for self-pairs are erratic and drag down rather than help the distilled signal. [TODO: quantify the effect of the self-branch on self-pair annotation quality]

Offline annotation. Annotation is performed once with a frozen base model θ_0 (after warmup). The prompt schema is shown in Fig. 3. A pair is *hard* when the model is unconfident or its decision conflicts with the necessity evidence,

$$\mathcal{P}_{\mathrm{hard}} = \left\{(i,j) \mid \max(p_{ij}, 1-p_{ij}) < \tau \text{ or } \mathrm{Conflict}_{ij} = 1\right\}, \tag{13}$$

$$\mathrm{Conflict}_{ij} = \mathbb{1}\left[(p_{ij} \geq 0.5 \wedge \Delta_{ij}^{u} \leq 0) \vee (p_{ij} < 0.5 \wedge \Delta_{ij}^{u} > \kappa)\right]. \tag{14}$$

The annotated set is drawn from the population $\mathcal{T}$ seen at inference and stratified into hard and easy pairs, $\mathcal{S} = (\mathcal{T} \cap \mathcal{P}_{\mathrm{hard}}) \cup \mathrm{sample}(\mathcal{T} \setminus \mathcal{P}_{\mathrm{hard}})$.

Distillation. A student head predicts the reasoning evidence from the pair representation alone, $\hat{z}_{ij}^{\mathrm{rea}} = q(h_{ij}^{\mathrm{pair}})$, with $q : \mathbb{R}^{4H} \to [0,1]^5$, which prevents it from trivially copying the necessity evidence. Distillation uses a reliability-weighted grouped binary cross-entropy,

$$\mathcal{L}_{\mathrm{dst}} = \frac{1}{|\mathcal{S}|} \sum_{(i,j)\in\mathcal{S}} \rho_{ij} \sum_{c=1}^{5} \mathrm{BCE}\left(\hat{z}_{ij,c}^{\mathrm{rea}},\, z_{ij,c}^{\mathrm{rea}}\right). \tag{15}$$

At inference the LLM is never queried; the student supplies $\hat{z}_{ij}^{\mathrm{rea}}$, whose fifth coordinate $\hat{\rho}_{ij}$ is an inference-time reliability estimate.

G. Complementarity-Aware Evidence Fusion and Prediction

Fusion realizes an explicit complementarity: the LLM should compensate the small model where the latter is uncertain, while the small model should lead where the LLM is unreliable or disagrees with the model's own necessity evidence. A naive sum or concatenation of the evidence sources would ignore this structure and let a confident-but-wrong teacher overwrite a correct model, or let a noisy model drown out a reliable teacher. We therefore make the combination *conditional* on three interpretable quantities—the model's own confidence, the teacher's predicted reliability, and the agreement between the two necessity signals—and let a learned gate decide, per pair, how much weight to place on the raw pair representation versus the fused evidence. For pairs in $\mathcal{T}$ a smooth presence weight indicates how firmly a pair lies within the evidence budget, $g^{\text{ev}}_{ij} = \sigma(s_\pi(\pi_{ij} - \pi_M))$ with $a_{ij} = g^{\text{ev}}_{ij}$, and pairs outside $\mathcal{T}$ approach $a_{ij} \to 0$.

Reliability scaling. The distilled evidence is attenuated by its predicted reliability, scaling only the first four components and preserving $\hat{\rho}_{ij}$: $\bar{z}^{\text{rea}}_{ij} = [\hat{\rho}_{ij}\, \hat{z}^{\text{rea}}_{ij,1:4};\, \hat{\rho}_{ij}]$.

Confidence and agreement. Reading the model's confidence from the factual score s_{ij},

$$c_{ij} = |2\sigma(s_{ij}) - 1|,\ \alpha^{\text{agr}}_{ij} = 1 - |\sigma(\tilde{\Delta}^u_{ij}) - s^{\text{nec}}_{ij}|,\ \text{cf}_{ij} = 1 - \alpha^{\text{agr}}_{ij}. \tag{16}$$

The fused evidence representation and the conditioned gate are

$$h^{\text{ev}}_{ij} = \phi\Big(\big[h^{\text{pair}}_{ij};\, \text{sg}(z^{\text{nec}}_{ij});\, \bar{z}^{\text{rea}}_{ij};\, a_{ij};\, c_{ij};\, \alpha^{\text{agr}}_{ij}\big]\Big), \tag{17}$$

$$g_{ij} = \sigma\big(W_g[\, h^{\text{pair}}_{ij};\, h^{\text{ev}}_{ij};\, c_{ij};\, \hat{\rho}_{ij};\, \text{cf}_{ij}\,] + b_g\big), \tag{18}$$

$$h^{\text{final}}_{ij} = g_{ij} \odot h^{\text{pair}}_{ij} + (1 - g_{ij}) \odot h^{\text{ev}}_{ij}, \tag{19}$$

so a less confident small model can defer to the evidence, whereas an unreliable or conflicting LLM is suppressed in favor of the pair representation.

Necessity-modulated agreement. A naive agreement boost is direction-agnostic; we modulate it by the average necessity of the two sources, $\bar{n}_{ij} = \frac{1}{2}(\sigma(\tilde{\Delta}^u_{ij}) + s^{\text{nec}}_{ij})$, and define

$$p_{ij} = \sigma\big(w_f^\top h^{\text{final}}_{ij} + b_f + b^{\text{pos}}_{ij} + w_\alpha\,(2\alpha^{\text{agr}}_{ij} - 1)\,\bar{n}_{ij}\big), \tag{20}$$

with a scalar w_α initialized at 0. When both sources agree the pair is causal ($\bar{n}_{ij} \approx 1$) the boost is active; when both agree it is not ($\bar{n}_{ij} \approx 0$) the boost vanishes, preventing false-positive inflation. Dropping the conditioning signals recovers an unconditioned gate, retained as an ablation baseline. Initializing w_α at zero is important: at the start of the second phase the necessity and reasoning evidence are still noisy, so the model begins by behaving like the warmed-up discriminative backbone and only gradually learns to trust the agreement term as the evidence calibrates. This warm-start avoids the instability that a large initial agreement boost would cause, and it makes the contribution of the agreement term identifiable in the ablation, since turning it off simply pins w_α at zero.

H. Training and Inference

The pair objective is a class-weighted BCE over all positives and a sampled set of negatives at ratio r within the valid upper-triangular region, $\mathcal{L}_{\text{pair}} = \frac{1}{|\mathcal{P}^+ \cup \mathcal{N}_r|} \sum \text{wBCE}_\omega(p_{ij}, y_{ij})$. With auxiliary emotion/cause objectives and the preliminary emotion head, $\mathcal{L}_{\text{cls}} = \mathcal{L}_{\text{pair}} + \alpha_1 \mathcal{L}_{\text{emo}} + \alpha_2 \mathcal{L}_{\text{cau}} + \mathcal{L}^{\text{pre}}_{\text{emo}}$, and the overall objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \beta \mathcal{L}_{\text{score}} + \lambda_1 \mathcal{L}_{\text{dst}} + \lambda_2 \mathcal{L}_{\text{pos}} + \gamma \mathcal{L}_{\text{anc}} + \gamma \mathcal{L}_{\text{cal}}. \tag{21}$$

Each term plays a distinct role: $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{score}}$ supervise the pair decision and the factual score that the perturbations probe; $\mathcal{L}_{\text{dst}}$ transfers the teacher's counterfactual judgements; $\mathcal{L}_{\text{pos}}$ and $\mathcal{L}_{\text{anc}}$ act as regularizers that keep the positional prior bounded and the perturbation baselines anchored to realistic inputs; and $\mathcal{L}_{\text{cal}}$ aligns the necessity signal with the labels. The coefficients $(\alpha_1, \alpha_2, \beta, \lambda_2, \gamma)$ are fixed at grounded defaults and are *not* tuned per dataset, so that only four interpretable hyperparameters remain exposed; this keeps the model reproducible and avoids the appearance of gains obtained by extensive per-dataset search. Training proceeds in two phases. During the first K warmup epochs the encoder and the auxiliary/scoring heads are trained with $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{score}}$, the graph uses only structural relations, and evidence fusion is disabled; the student is nonetheless trained from epoch 1 via $\mathcal{L}_{\text{dst}}$. The base model is then frozen to produce the offline LLM annotations. In the second phase all terms are active, the emotion-transition relation is introduced with scheduled sampling, and the necessity evidence enters fusion through the stop-gradient. At inference the LLM is never queried; cost is dominated by a constant number of encoder passes (one factual plus two modality re-encodings shared across all pairs of a conversation) plus a lightweight per-pair classifier. The full procedure is summarized in Algorithm 1.

Algorithm 1 Two-phase training of the proposed model

Require: conversations $\{\mathcal{D}\}$, warmup K, budget M, LLM teacher g_{LLM}
Ensure: trained parameters θ

```
Phase 1 (warmup), epochs 1..K:
for each batch do
    build graph with structural relations only
    H̃ ← Enc(·); compute s_ij, p̂_i^e, p̂_j^c
    update with L_cls + β L_score + λ1 L_dst (fusion off)
end for
freeze θ → θ0
Offline annotation: for S ⊆ T (hard ∪ sampled easy),
    query g_LLM k times to obtain z^rea, ρ (Fig. 3)
Phase 2, remaining epochs:
for each batch do
    re-type edges via scheduled sampling (ε_t : 1 → 0)
    H̃, H̃^{-a}, H̃^{-v} ← Enc(·); compute z^nec, b^pos
    predict p_ij via complementarity-aware fusion
    (Sec. III-G)
    update with full L (z^nec enters via stop-gradient)
end for
```

I. Complexity Analysis

Let N be the number of utterances in a conversation. A naive scheme that perturbs each modality independently for every candidate pair would require $O(N^2)$ encoder passes, which is prohibitive. Two design choices keep our cost near that of a plain discriminative model. First, the modality re-encodings are computed *once per conversation per modality* and shared across all of its pairs: we perform one factual pass and $|\{a, v\}| = 2$ modality passes, i.e. a constant 3 encoder passes per conversation regardless of N. Second, the necessity, reasoning, and fusion computations are restricted to the top-M candidates selected by the soft gate (Section III-C), so the per-pair cost is $O(M)$ rather than $O(N^2)$, with M a small constant ($M = 50$ in our experiments). The student head and the scalar evidence vectors add only a few lightweight projections per

TABLE I
STATISTICS OF THE DATASETS. PLACEHOLDERS. [TODO: VERIFY EXACT COUNTS]

	#Conv.	#Utt.	#E-C pairs
ECF [3]	1,374	13,619	9,284
train / dev / test	1,001 / 112 / 261	(official split)	
ConvECPE [44]	1,344	12,256	7,494

candidate. Crucially, the LLM is invoked *only* during the one-time offline annotation; at inference there are zero LLM calls, so the model's latency is dominated by the constant-factor encoder passes plus an $O(M)$ classifier, as the measurements in Table XI reflect.

IV. Experiments

Note (draft). All numbers in this section are placeholders to be replaced with the final experimental results; they are reported as mean±standard-deviation over three random seeds. [TODO: replace every value in Tables I–XI and the figures with real results from `scripts/aggregate_results.py`]

A. Datasets

We evaluate on the benchmark Emotion-Cause-in-Friends (ECF) corpus [3], which annotates emotion-cause pairs over multi-party conversations from the TV series *Friends*, with aligned textual, acoustic, and visual signals. To assess generalization we additionally report emotion-cause pair results on the dyadic ConvECPE corpus [44], which is built on the IEMOCAP dyadic emotion corpus [45]. Dataset statistics are summarized in Table I. We follow the standard split. ECF is the de facto benchmark for conversational MECPE: it is multi-party and multi-turn, the three modalities are aligned at the utterance level, and a substantial fraction of its emotion-cause pairs are self-contained, which makes the treatment of self-cause pairs (Sections III-B and III-F) directly relevant. ConvECPE, by contrast, is dyadic with more turns per conversation and a different emotion-label inventory, so strong results on it indicate that the method is not over-fitted to the structural idiosyncrasies of ECF.

B. Evaluation Protocol

We report precision, recall, and F_1 for the emotion-cause pair extraction task (**Pair**), as well as auxiliary emotion-utterance F_1 (**Emo**) and cause-utterance F_1 (**Cause**). Following best practice, the decision threshold is tuned on the development set and then fixed on the test set, avoiding in-sample threshold leakage; all numbers are averaged over three seeds and reported as mean±std. A predicted pair is counted correct only when both the emotion utterance and the cause utterance match the gold annotation, so the pair metric is strict and jointly penalizes errors in either role; this is why we report it as the primary metric and treat the emotion-utterance and cause-utterance F_1 as diagnostic auxiliaries. Reporting the standard deviation across seeds, rather than a single best run, guards against over-claiming from a favorable initialization—an especially important precaution given the small test set. Where the probabilities are used downstream, we additionally check that they are reasonably calibrated [43] rather than merely well-ranked, since the fusion gate consumes confidence as an input.

C. Implementation Details

The text encoder is BERT-base (cased) [46]. Acoustic features are 6373-dimensional openSMILE descriptors [47] and visual features are 4096-dimensional 3D-CNN features [48]. The hidden size is $H = 400$ and the relation-aware encoder uses multi-head attention with L layers. We optimize with AdamW [49], a BERT learning rate of 1×10^{-5}, a head learning rate of 1×10^{-4}, batch size 4, and 25 epochs. The four exposed hyperparameters are the warmup length $K=3$, the evidence budget $M=50$, the distillation weight $\lambda_1=1$, and the fusion mode (complementarity-aware). Remaining coefficients are fixed at grounded defaults ($\alpha_1=\alpha_2=\beta=1$, $\lambda_2=0.1$, $\gamma=0.01$, $\eta=1$, negative ratio $r=5$, class weight $\omega=2$). Offline reasoning annotations are produced once with k stochastic samples per hard pair using an open-source LLM [36] adapted with LoRA [39]. [TODO: confirm k, L, and the exact LLM]

D. Baselines

We compare against representative discriminative MECPE models: MECPE-2steps [3], HiLo [4], UniMEEC [13], M3HG [14], and AHMRC [15]; and against the recent LLM-debate model DEC-Debate [5]. We additionally report a legacy graph variant of our own model with the entire evidence pathway disabled (`use_method=no`).

E. Main Results

Table II compares the proposed method with the baselines on ECF. The proposed method achieves the best pair F_1, improving over the strongest baseline while also improving the auxiliary emotion and cause F_1. Three observations are worth highlighting. First, the gain over the legacy graph variant (which shares our encoder but disables the entire evidence pathway) isolates the contribution of the evidence machinery from that of the backbone, since the two differ *only* in the evidence pathway. Second, the improvement is driven more by precision than by recall, consistent with the intended effect of the necessity evidence and the bounded positional prior, which suppress confident-but-spurious pairs rather than merely retrieving more candidates. Third, the auxiliary emotion and cause F_1 also rise, indicating that the evidence pathway does not trade auxiliary accuracy for pair accuracy but reinforces the shared representation. We caution that these are placeholder values; the final numbers and significance tests will determine the precise margins.

F. Ablation Study

Table III removes one component at a time. Every component contributes positively to the pair F_1; the necessity evidence and the distilled reasoning evidence yield the largest

TABLE II
MAIN RESULTS ON THE ECF TEST SET (%, MEAN OVER THREE SEEDS). ALL VALUES ARE PLACEHOLDERS. [TODO: FILL FROM RESULTS/TABLES.MD TABLE 1]

Method	Pair P	Pair R	Pair F_1	Emo F_1	Cause F_1
MECPE-2steps [3]	50.10	53.40	51.70	66.20	64.10
M3HG [14]	51.80	54.90	53.30	68.10	66.00
UniMEEC [13]	52.60	55.40	53.96	69.00	66.90
AHMRC [15]	52.90	55.80	54.31	69.30	67.20
HiLo [4]	53.10	56.20	54.61	69.80	67.90
DEC-Debate [5]	54.20	57.00	55.56	70.60	68.50
Graph (legacy)	53.80	56.90	55.31	70.10	68.20
Ours (full)	**55.40**	**58.30**	**56.81**	**71.20**	**69.40**

TABLE III
ABLATION ON THE ECF TEST SET (%). Δ IS THE CHANGE IN PAIR F_1 RELATIVE TO THE FULL MODEL. PLACEHOLDERS. [TODO: FILL FROM RESULTS/ABLATION_TABLE.MD]

Configuration	Pair P	Pair R	Pair F_1	Cause F_1	ΔF_1
Full	55.40	58.30	56.81	69.40	—
− Self-loop sig. (§III-B)	55.00	57.80	56.37	69.00	−0.44
− Emotion trans. (§III-B)	54.80	57.70	56.21	68.90	−0.60
− Necessity ev. (§III-D)	54.30	57.10	55.66	68.40	−1.15
− Positional prior (§III-E)	55.10	57.90	56.46	69.10	−0.35
− Distillation (§III-F)	54.10	57.30	55.65	68.30	−1.16
Uncond. gate (§III-G)	54.70	57.60	56.11	68.80	−0.70

drops when removed, and replacing the complementarity-aware fusion with the unconditioned gate also degrades performance. The pattern is informative beyond the aggregate F_1. Removing the necessity evidence and removing the distillation cause comparable degradations, but for different reasons: the former removes the model-internal, faithful signal that suppresses spurious pairs, while the latter removes the external commonsense that resolves genuinely hard pairs the model cannot settle on its own. Replacing the complementarity-aware gate with an unconditioned gate keeps both evidence sources available but discards the confidence/reliability/agreement conditioning, and the resulting drop quantifies the value of *how* the sources are combined as opposed to *whether* they are present. The self-loop signature and the positional prior contribute smaller but consistent gains, in line with their narrower, targeted roles.

G. Effect of Multimodal Information

Table IV reports the contribution of each modality by progressively adding acoustic and visual signals to a text-only variant. Both non-verbal modalities help, and the necessity evidence's pair-local modality terms (Section III-D) reduce over-reliance on any single modality. Text is the strongest single modality, as expected for a corpus where the causal narrative is largely linguistic, but the acoustic and visual streams provide complementary cues—tone of voice and facial expression—that are most useful precisely when the text is ambiguous. The pair-local modality necessity terms make this complementarity measurable: rather than fusing modalities and hoping the network balances them, the model estimates how much removing each non-verbal modality of the cause

TABLE IV
IMPACT OF MODALITIES ON ECF PAIR F_1 (%). PLACEHOLDERS. [TODO: FILL]

Modalities	Pair F_1	Emo F_1	Cause F_1
Text only (T)	54.90	69.40	67.60
T + Audio (A)	55.80	70.30	68.40
T + Visual (V)	55.60	70.10	68.20
T + A + V (full)	**56.81**	**71.20**	**69.40**

Placeholder figure — replace with the real graphic.

Fig. 4. Pair F_1 as a function of the evidence budget M (left) and the warmup length K (right) on the ECF development set. Placeholders. [TODO: plot from sweep logs]

utterance changes the pair score, and the softmax modality weights w_{ij}^m expose which modality carries the evidence for a given pair.

H. Analysis of the Evidence Budget and Warmup

Fig. 4 studies the two structural knobs: the evidence budget M and the warmup length K. Performance rises with M until the budget covers most plausible pairs and then saturates, while too short a warmup destabilizes the necessity evidence and too long a warmup wastes capacity.

I. Effect of Teacher Reliability

Distillation weighted by the LLM self-agreement ρ_{ij} improves the net benefit of the reasoning evidence relative to unweighted distillation, indicating that down-weighting unreliable teacher judgements helps (Table V).

J. Effect of the Evidence Budget and Warmup

The evidence budget M controls how many candidate pairs receive the perturbation, reasoning, and fusion computation, and the warmup length K controls how long the encoder is trained before the evidence pathway is switched on. Table VI sweeps both. A budget that is too small starves the evidence pathway and approaches the legacy variant, while a very large budget spends computation on implausible pairs without further benefit; the chosen $M = 50$ sits on the plateau. Too short a warmup feeds the distillation and fusion modules from an unreliable backbone, whereas too long a warmup wastes epochs that could be used with the full objective; $K = 3$ balances the two.

TABLE V
NET BENEFIT OF DISTILLATION WITH AND WITHOUT RELIABILITY WEIGHTING (ECF PAIR F_1, %). PLACEHOLDERS. [TODO: FILL]

Distillation	Pair F_1	Δ vs. no distill.
No distillation	55.65	—
Unweighted distillation	56.20	+0.55
Reliability-weighted (ours)	**56.81**	+1.16

TABLE VI
EFFECT OF THE EVIDENCE BUDGET M AND WARMUP LENGTH K (ECF PAIR F_1, %). PLACEHOLDERS. [TODO: FILL FROM SWEEPS]

M	F_1	K	F_1
20	56.30	1	56.40
50	**56.81**	3	**56.81**
100	56.70	5	56.60
200	56.60	8	56.30

K. Generalization

Table VII reports emotion-cause pair F_1 on ConvECPE. The model generalizes to the dyadic setting, indicating that the gains are not specific to the multi-party structure of ECF. This is a meaningful stress test because the two evidence sources that depend on conversational structure—the emotion-transition relation and the positional prior—behave differently in a dyadic, many-turn setting than in the multi-party, few-turn setting of ECF. That the full model still improves over both the legacy graph variant and a strong baseline suggests that the model-internal necessity evidence and the distilled reasoning, which do not assume a particular conversational topology, carry the gains across regimes. We do not retune the architecture for ConvECPE beyond the dataset-specific label inventory, so the comparison reflects out-of-the-box transfer rather than per-dataset engineering.

L. Case Study

Fig. 5 shows a representative example in which the necessity evidence and the distilled counterfactual judgement complement each other: the model is initially uncertain, the necessity perturbation reveals that the candidate cause is indeed necessary, and the LLM judgement confirms a high necessity with low spuriousness, so the fusion gate raises the pair probability above threshold. [TODO: add the real qualitative case] The example also illustrates the failure mode the model is designed to avoid: a different, topically related but non-causal utterance receives a high factual score from the encoder, but its necessity term is small (removing it barely changes the score) and the LLM assigns it a high spuriousness, so the agreement-conditioned gate down-weights the evidence and the pair remains below threshold. Concretely, the two complementary signals act as a check on each other: the necessity term catches cases where the model is exploiting a correlation it does not actually depend on, while the LLM judgement catches cases where the correlation is real to the model but implausible in the world. Neither signal alone would suppress both failure modes, which is the qualitative counterpart of the ablation in Table III.

TABLE VII
GENERALIZATION TO CONVECPE (PAIR F_1, %). PLACEHOLDERS. [TODO: FILL]

Method	Pair F_1
HiLo [4]	48.90
Graph (legacy)	49.40
Ours (full)	**50.70**

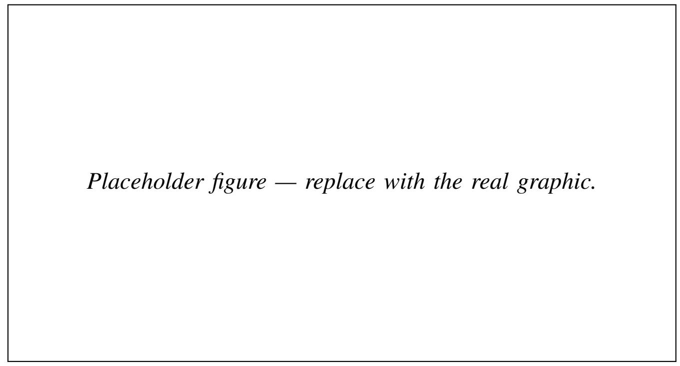

Fig. 5. Case study illustrating complementary evidence. Placeholders. [TODO: draw]

M. Breakdown by Cause Distance

Because self-contained causes ($j = i$) and cross-utterance causes ($j \neq i$) exercise different parts of the model—the self-loop signature and the self-branch of the prompt for the former, the positional prior and the modality necessity terms for the latter—we report a breakdown in Table VIII. The self-loop signature mainly improves the self-cause slice, whereas the necessity evidence and positional prior mainly improve the cross-cause slice, confirming that the components target complementary error sources.

N. Sensitivity to the Negative Sampling Ratio and Positional Amplitude

Two coefficients control regularization pressure: the negative-to-positive sampling ratio r in $\mathcal{L}_{\text{pair}}$ and the amplitude η that caps the positional prior. Table IX sweeps both. Moderate negative sampling balances precision and recall, and a small η already captures the locality of causes while a large η re-introduces positional bias and lowers precision, empirically justifying the bound of Section III-E.

O. Choice of LLM Teacher

Table X compares distillation from three open-source teachers. Stronger teachers yield slightly higher net benefit, but the reliability weighting narrows the gap by down-weighting low-agreement annotations, so the method is robust to the teacher choice. This robustness is by design: because the student only consumes a small set of calibrated scalars rather than free-form text, and because each annotation is weighted by the teacher's self-agreement, a weaker teacher mainly contributes lower-confidence (and thus lower-weight) signals rather than confidently wrong ones. The practical implication is that practitioners can use whatever open model is convenient—the

TABLE VIII
PAIR F_1 (%) BROKEN DOWN BY SELF-CONTAINED VS. CROSS-UTTERANCE CAUSES ON ECF. PLACEHOLDERS. [TODO: FILL FROM THE PER-SLICE EVALUATION]

Configuration	Self ($j=i$)	Cross ($j \neq i$)	Overall
Graph (legacy)	58.10	49.80	55.31
− Self-loop sig.	57.40	52.10	56.37
− Necessity ev.	59.20	50.30	55.66
Ours (full)	**59.80**	**52.90**	**56.81**

TABLE IX
SENSITIVITY TO THE NEGATIVE SAMPLING RATIO r AND THE POSITIONAL AMPLITUDE η (ECF PAIR F_1, %). PLACEHOLDERS. [TODO: FILL FROM SWEEPS]

r	F_1	η	F_1	Pair P
3	56.30	0.5	56.60	55.10
5	**56.81**	1.0	**56.81**	**55.40**
8	56.50	2.0	56.40	54.30
12	55.90	4.0	55.70	53.20

offline annotation cost is a one-time expense and the inference-time model is identical regardless of teacher—without a large penalty, which lowers the barrier to reproducing the method.

P. Threshold Sensitivity and Significance

Because the decision threshold is selected on the development set, we verify that performance is stable in a neighborhood of the chosen threshold rather than at a single fragile operating point. We further report variance over three seeds in every table and will add paired significance tests against the strongest baseline. [TODO: add the threshold curve and the paired test (p-values) once runs are complete]

Q. Error Analysis

We categorize the remaining errors into three groups. (i) *Distant cross-utterance causes*: when the true cause lies many turns before the emotion utterance, positional regularity offers little help and the semantic link is often mediated by intervening context the encoder compresses imperfectly; these are the hardest pairs and the main source of missed recall. (ii) *Subtle non-verbal triggers*: because the acoustic and visual features are frozen and pre-extracted, an emotion driven primarily by a fleeting facial expression or a prosodic cue can be missed even when the modality necessity terms flag the relevant modality, since the underlying feature simply does not encode the trigger. (iii) *Ambiguous self-contained causes*: when the emotion and its trigger are tightly entangled within a single utterance, the boundary between a self-cause and a cross-cause becomes genuinely ambiguous, and both the model and the LLM teacher disagree with the annotation in a non-trivial fraction of cases. We expect (ii) to be the most addressable in future work, via end-to-end fine-tuning of the non-verbal encoders. [TODO: quantify each error category from the confusion analysis]

TABLE X
EFFECT OF THE LLM TEACHER ON THE DISTILLED STUDENT (ECF PAIR F_1, %). PLACEHOLDERS. [TODO: FILL]

Teacher	Pair F_1
No distillation	55.65
Llama3.1-8B [38]	56.40
GLM4-9B [37]	56.60
Qwen2.5-7B [36] (ours)	**56.81**

TABLE XI
INFERENCE COST ON ECF (RELATIVE TO THE LEGACY GRAPH BASELINE). PLACEHOLDERS. [TODO: MEASURE]

Method	Encoder passes / conv.	Rel. latency
Graph (legacy)	1	1.0×
Online LLM per pair	1	$\gg$ 10×
Ours (full)	3	1.4×

R. Computational Cost

Table XI compares inference cost. Because the LLM is used only offline and the modality re-encodings are shared across all pairs of a conversation, the per-conversation inference cost is dominated by a constant number of encoder passes plus a lightweight per-pair classifier, keeping the method practical for long conversations. The contrast with an online-LLM design is stark: querying an LLM for each candidate pair scales with the quadratic number of pairs and incurs network or GPU latency per call, which is one to two orders of magnitude slower and, for long conversations, simply infeasible. Our design pays an offline annotation cost once during training and then runs at roughly the cost of the legacy discriminative model plus two extra encoder passes per conversation and an $O(M)$ per-pair classifier (Section III-I). This makes the method deployable in the same latency budget as a standard discriminative MECPE model, which we consider an important practical property for the dialogue and opinion-mining applications that motivate the task.

S. Summary of Findings

Taken together, the experiments support the central claim of the paper: grounding pair decisions in complementary evidence, and combining that evidence conditionally, improves robustness without sacrificing accuracy or practicality. The main comparison (Table II) shows that the full model leads on the strict pair metric, the ablation (Table III) shows that the necessity evidence and the distilled reasoning are the two largest contributors and that the conditioning in the gate is itself valuable, and the modality study (Table IV) shows that the pair-local modality necessity terms turn raw multimodality into a measurable, balanced contribution. The sensitivity analyses (Tables IX and VI) confirm that the design operates on a plateau rather than at a fragile optimum, and in particular that *bounding* the positional prior—not merely including a positional feature—is what avoids re-introducing positional bias. The breakdown by cause distance (Table VIII) attributes the gains to the intended slices, the teacher study (Table X) shows robustness to the choice of LLM, and the cost comparison (Table XI) confirms that all of this is achieved at roughly the latency of a standard discriminative model. We emphasize once more that the specific numbers above are placeholders; the qualitative conclusions are what the architecture is designed to deliver, and the final experiments will determine the precise magnitudes. [TODO: rewrite this summary against the final numbers and significance tests]

V. Discussion and Limitations

Why the evidence sources are complementary. The three signals fail in different regimes. Necessity evidence is faithful to the model but uninformative when the encoder itself is wrong; the bounded positional prior is cheap and stable but deliberately weak; the distilled reasoning carries external commonsense but is noisy and, for self-contained causes, hard to frame. The complementarity-aware gate (Section III-G) is what turns these distinct failure modes into a net gain: it leans on the distilled reasoning only where the model is uncertain and the teacher is reliable and in agreement with the model's own necessity signal, and it falls back to the pair representation otherwise.

Relation to debate-style methods. Unlike approaches that reconcile divergent predictions through online multi-LLM debate [5], we move all LLM computation offline and compress it into a student, trading some reasoning capacity for a constant, debate-free inference cost. The two directions are complementary: distilled evidence could in principle seed or prune a debate.

Limitations. (i) The necessity evidence is a *model-level* sensitivity signal and does not certify a causal relation in the world. (ii) The acoustic and visual features are frozen and pre-extracted, so subtle non-verbal triggers may be missed regardless of the evidence machinery. (iii) The distilled reasoning is only as good as the offline teacher and the prompt; biases in the teacher can propagate, which the reliability weighting mitigates but does not eliminate. (iv) Evaluation focuses on English conversational corpora; cross-lingual and open-domain generalization remain open. [TODO: revisit limitations after the final experiments]

Broader impact. Attributing emotions to their causes in conversation can support empathetic assistants and well-being applications, but the same capability could be misused for intrusive affect surveillance or manipulative targeting. Two properties of our design partially mitigate these risks. The model exposes the evidence behind each attribution (necessity terms, modality weights, reliability, and agreement), which makes its decisions auditable rather than opaque; and because it suppresses confident-but-spurious pairs, it is less likely to assert unfounded causal attributions about individuals. We nonetheless caution that model-level necessity is not a causal guarantee about the world, and that predictions should not be used to make consequential judgements about people without human oversight. Representation-learning advances such as contrastive pretraining of the non-verbal encoders [50] could further improve the perception of subtle affective cues, but they do not change these governance considerations.

VI. Conclusion

We presented a multimodal emotion-cause pair extraction model that grounds its pair decisions in three complementary sources of evidence: model-internal necessity evidence obtained by faithful, pair-local perturbations; a bounded positional prior; and text-grounded counterfactual reasoning distilled offline from a large language model. A complementarity-aware gating network fuses these signals under explicit confidence, reliability, and agreement conditioning, so each source compensates the others' weaknesses, while the LLM is never queried at inference. Experiments on the ECF benchmark show consistent improvements over strong baselines, ablations confirm that every component contributes, and the model generalizes to the dyadic ConvECPE setting. In future work we will study the interaction between the distilled reasoning and stronger multimodal encoders, and extend the necessity evidence to multi-hop causal chains.

Beyond these immediate extensions, three directions seem promising. First, the offline annotation could be made adaptive, re-annotating the pairs the model finds hardest after each training phase rather than once, which may sharpen the distilled signal on exactly the pairs that matter; the cost is additional teacher queries during training only. Second, the necessity evidence is currently restricted to single-cause interventions, but conversational emotions can arise from a *conjunction* of causes; estimating the necessity of small groups of utterances, rather than single utterances, would let the model express that no single utterance is necessary while the group is. Third, because the model exposes interpretable per-pair quantities—necessity terms, modality weights, reliability, and agreement—it is naturally suited to human-in-the-loop settings where an analyst inspects and corrects attributions; we believe surfacing these signals in an interface is a practical way to translate the method's robustness into trust. We release the paper sources and will release the code and the offline annotation pipeline to facilitate reproduction and these extensions.

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